PORTFOLIO

WORKS BY CHEN JIAHAO

2023-2024



CHEN JIAHAO

Basic Information

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Email:152683800064@163.com

Education Background

Anhui University
Major:Bachelor of Fine Arts in Environmental Design

Honors&Awards

Silver Award of the 14th China Space Design Competition (Anhui Province)	07/2024
Honorable Mention of 2024 eVolo Skyscraper Competition	06/2024
Academic Excellence Scholarship for the 2022-2023 academic year	12/2023
Distinction Award of 2023 YD DESIGN AWARD	08/2023
Third Prize of Anhui University Students Environmental Design Competition 2022	04/2023
Outstanding Student Leader for the academic year 2021 to 2022	12/2022
Academic Excellence Scholarship for the 2021-2022 academic year	12/2022

INTERNSHIP EXPERIENCE

Zhejiang Huazhi Architectural Design Co. Ltd. | Designer assistant	07/2024 - 08/2024
Johnson Design Group Co. Ltd. | Designer assistant	01/2024 - 02/2024

EXTRACURRICULAR EXPERIENCE

EVOLO Skyscraper Race | Croup member	11/2023 - 01/2024
2023 Innovation and Entrepreneurship Training Programme for University Students | Core member	02/2023 - 06/2023
Zhouzhuang Ancient Town Study Tour Activity | Assistant	04/2022 - 06/2022
Exhibition and Charity Sale | Volunteer	03/2022 - 06/2022

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-Water efficient community housing project in Monterey

■ **04 RAINBOW LINE** 10/2024-11/2024
-Coastal flood control infrastructure

■ **05 THE MEMORY OF ALEPPO** 11/2023-01/2024
-To collect city memories

■ **06 OTHER WORKS**

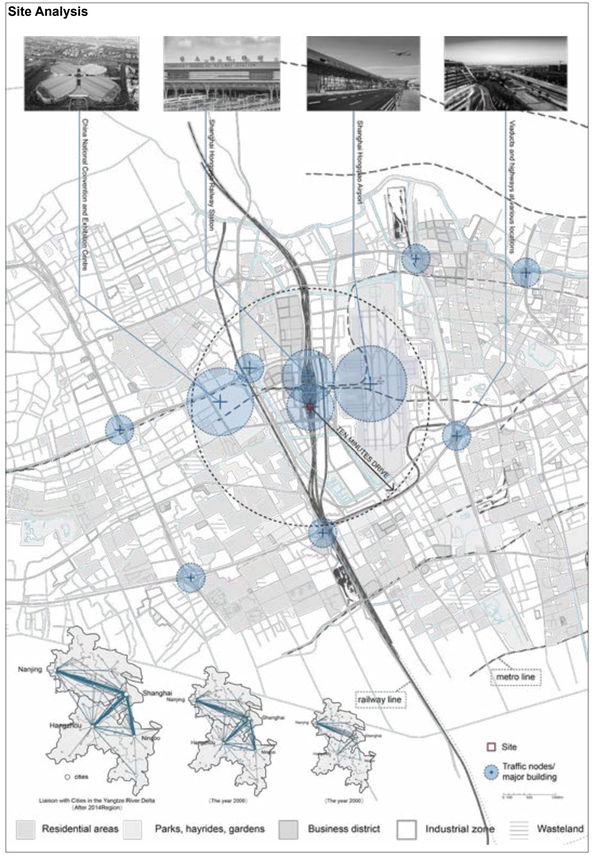


01 Urban wormhole -A complex above the Shanghai Railway

Type:Hotel/Community building
Date:2023.8-2024.1
individual work

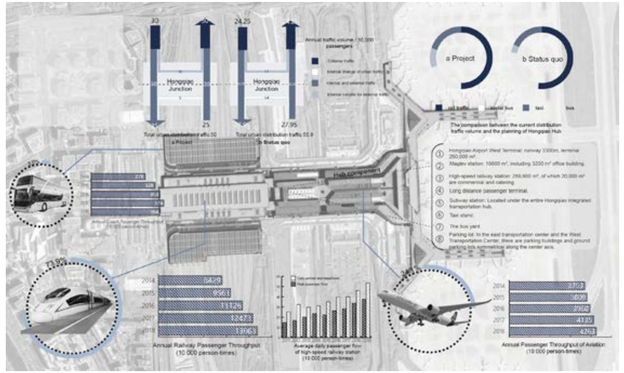
As a theoretical concept that connects different Spaces and times, cosmic wormholes provide infinite imagination for architectural design. Combining a cosmic wormhole with residential buildings and transportation hubs, the aim is to create a futuristic urban space that is creative, science-fiction-filled, and capable of efficient transportation and living. With the acceleration of the urbanization process in Shanghai, the population density continues to rise, and traffic congestion and space resource shortage have become urgent problems to be solved. At the same time, people are also curious about the living environment of exploring the unknown and pursuing science fiction. This project takes the universe wormhole as inspiration and integrates it into the design of residential buildings and transportation hubs. Through this design concept, we hope to provide a new possibility for urban development, and also bring people a better vision of future life.

Site Analysis

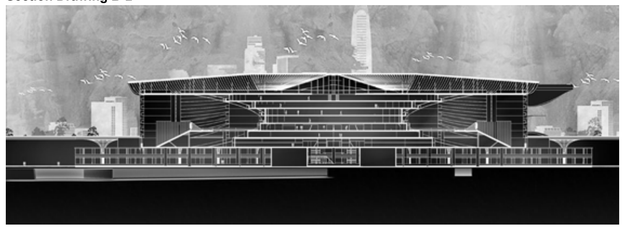




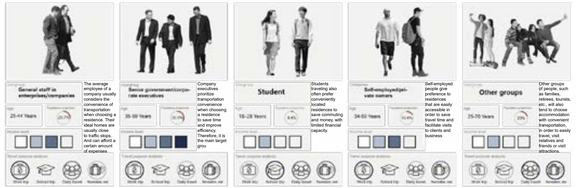
Passenger Analysis



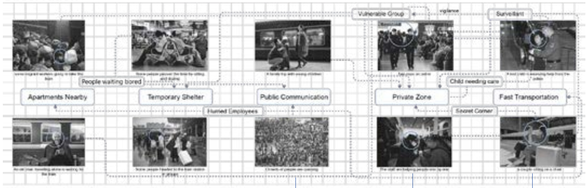
Section Drawing B-B'



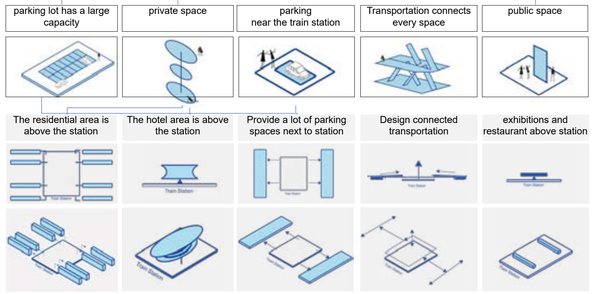
Customer Analysis



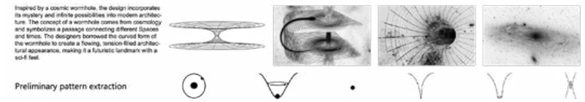
Observation



Strategy and Design Development



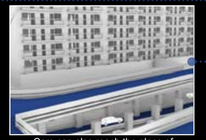
Design Inspiration

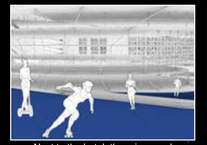


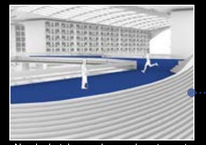


Function Perspective View


This place is a first floor park.Connected to the elevated road.


Cars can also reach the place of residence.

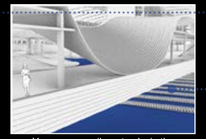

Next to the hotel, there is a road connecting the railway, which can directly reach the train station.


Nearby hotels can also reach restaurants and exhibition spaces.


The hotel also connects to the place of residence.


The bottom layer is the departure level of the high-speed train.

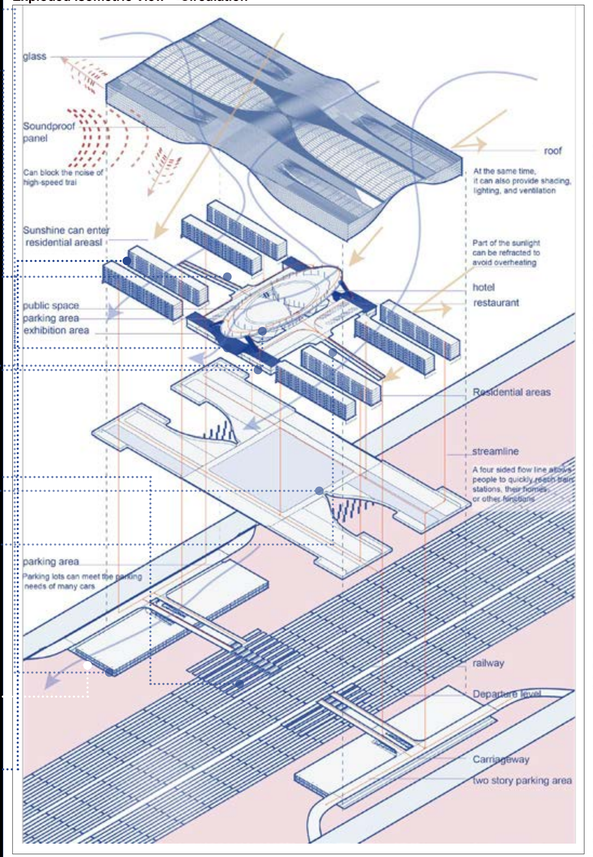

The car can also be driven up to the hotel.


You can see railway tracks in the restaurant and exhibition space.

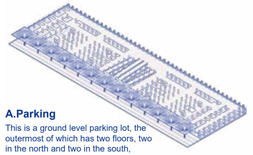

The parking lot has several floors to meet parking needs.


In residential areas, one can also enjoy privacy and tranquility.

Exploded Isometric View -Circulation

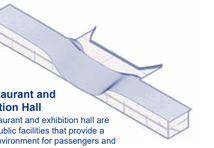


Functional Analysis



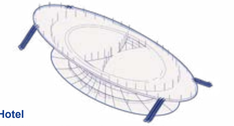
A.Parking

This is a ground level parking lot, the outermost of which has two floors, two in the north and two in the south, Under the elevated road, you can directly reach the train station



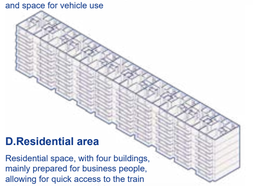
B.Restaurant and Exhibition Hall

The restaurant and exhibition hall are mainly public facilities that provide a dining environment for passengers and guests, with two in each direction.



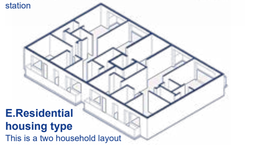
C.Hotel

Hotel facilities, providing guest room services for short-term stays.At the top is a pedestrian rooftop and space for vehicle use



D.Residential area

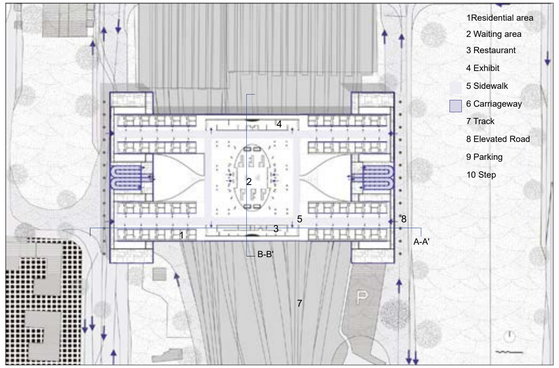
Residential space, with four buildings, mainly prepared for business people, allowing for quick access to the train station



E.Residential housing type

This is a two household layout with independent balconies, suitable for single and family accommodation.

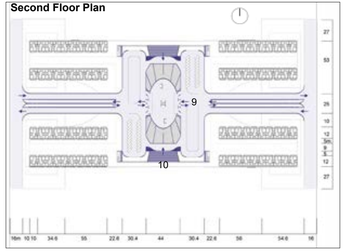
Ground Floor Plan



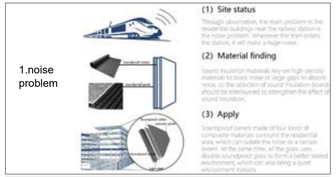
Parking Area Plan

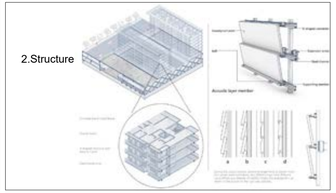


Second Floor Plan

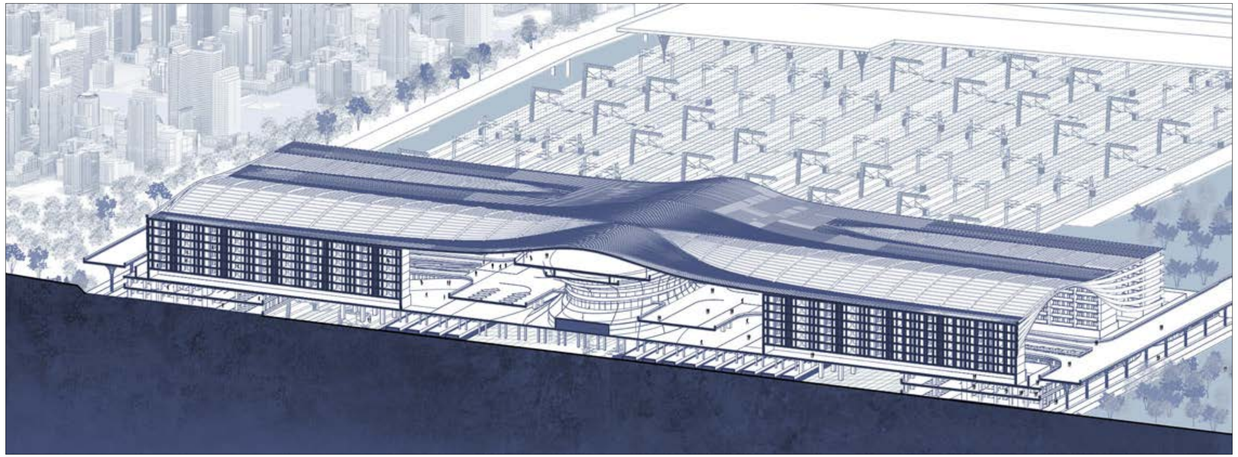


Detail Design





Section Drawing A-A'



Rendering



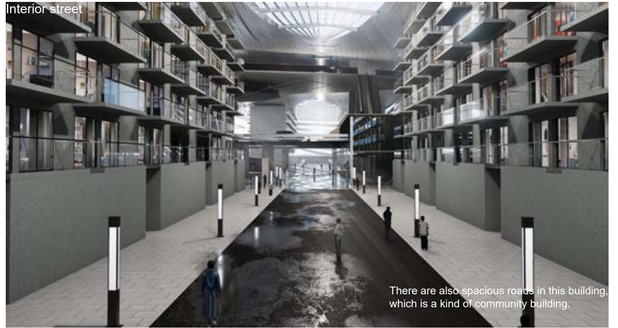
The city wormhole is adjacent to the elevated, and residents are extremely convenient to travel within the city or across provinces.
View from the street,showing the architectural panorama and the train passing through the scene.

Public park



It is a public rest place where people can relax and chat, and at the same time, they can have fun with their families and enjoy the beauty of the city.

Interior street



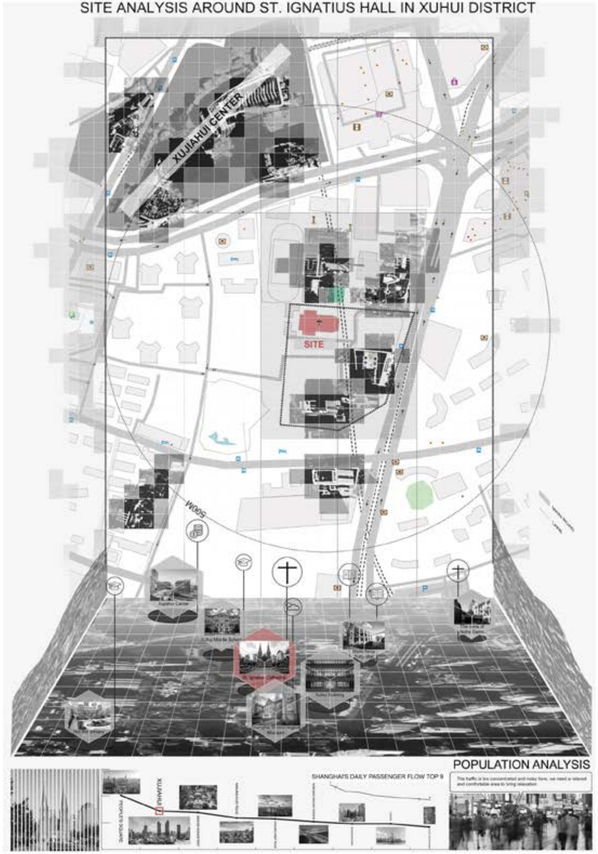
There are also spacious roads in this building, which is a kind of community building.

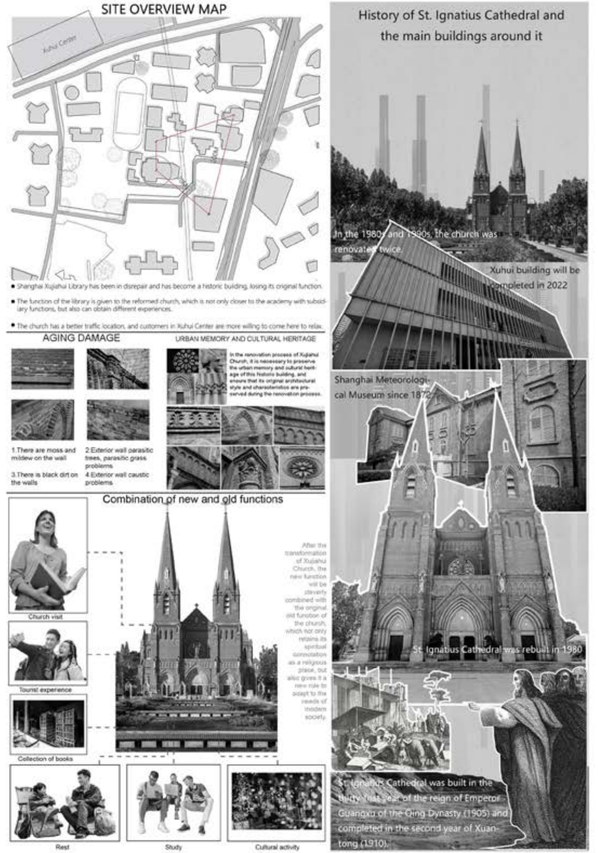


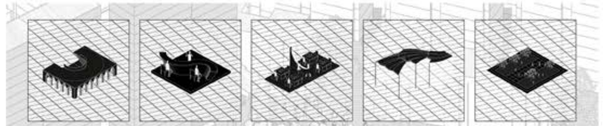
02 Paradise Library -Church renovation

Type:Renovation project
Date:2024.6-2024.8
individual work

In this renovation design of the church, I combined the traditional concept of a library museum with the church space, injecting new vitality and functions into this historic building. Respecting the original structure and historical value of the building, careful expansion and restoration work has been carried out to transform the church into a comprehensive cultural Hub, combining the functions of a library, museum, restaurant and shop.
The renovated church will not only become a space for book collection and display, but also play an important role in displaying local culture and history. In addition, the new dining and retail facilities will provide visitors with convenient places to rest and shop. Through this multi-functional design, we hope to attract wider public participation, promote knowledge dissemination and cultural exchange, and create a destination for local residents and visitors that integrates knowledge, culture, food and shopping, thereby adding rich content and vitality to community life.



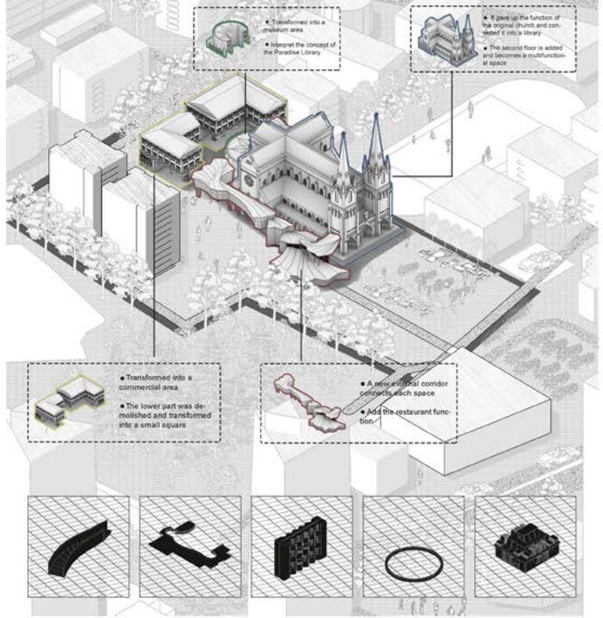


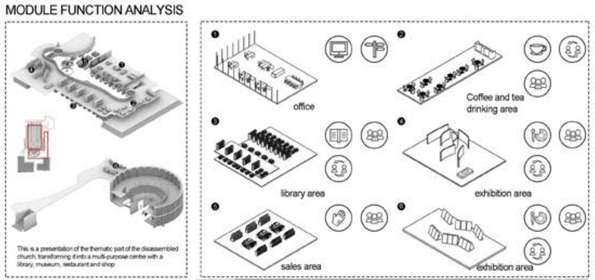


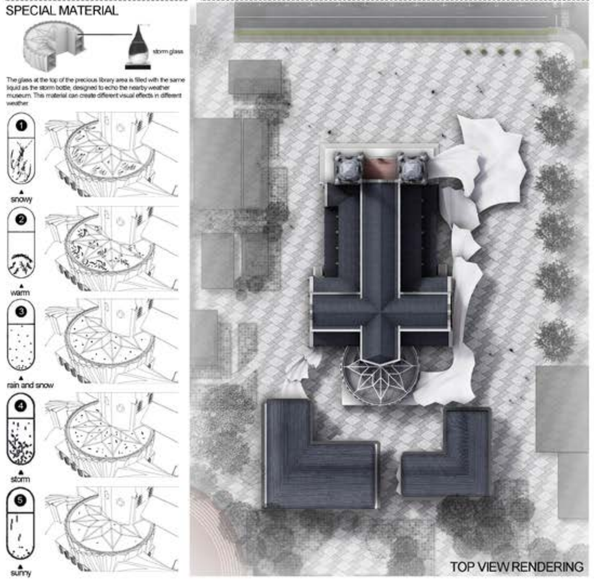
PARADISE LIBRARY

The design is inspired by the famous idea of Argentinian literary master Borges: "I have always secretly imagined that heaven should be like a library." This inspiraton led me to integrate the concept of the ancient collection museum into the renovation of the church, giving it new life and function. During the expansion and restoration process, the church will be transformed into a multi-purpose center with a library, museum, restaurant and shop, providing a comprehensive place for knowledge, culture, food and shopping.

FEATURE



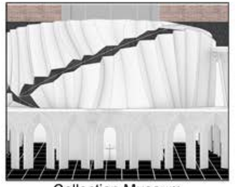


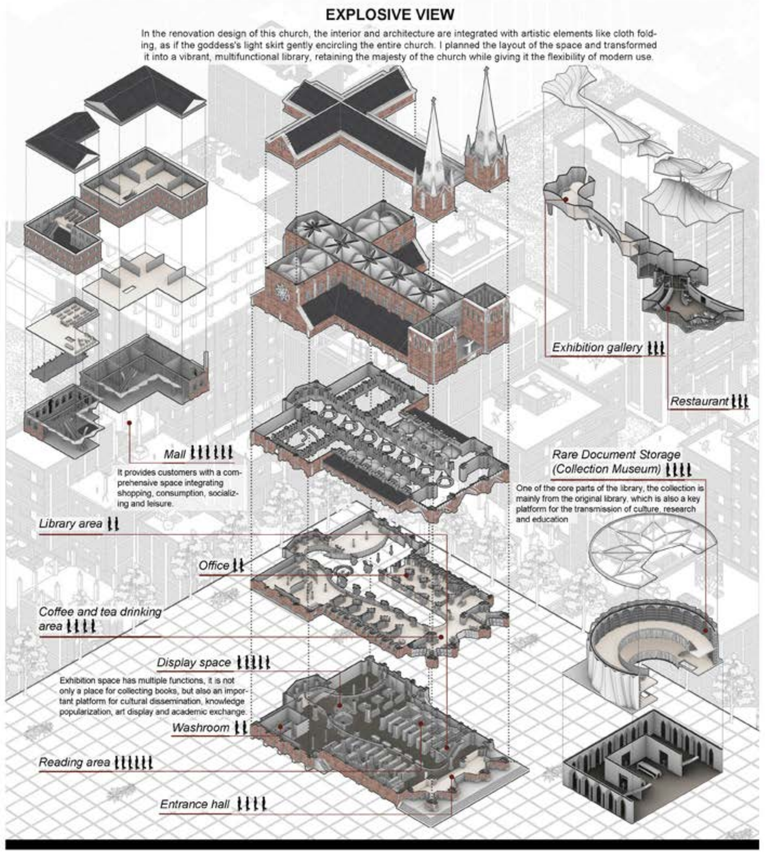



Restaurant

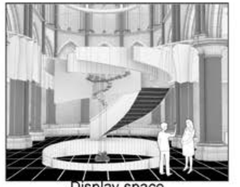

Office


Mall


Collection Museum

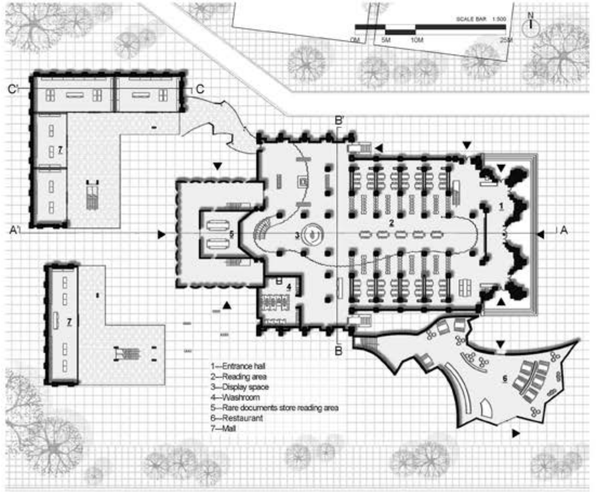



Reading area

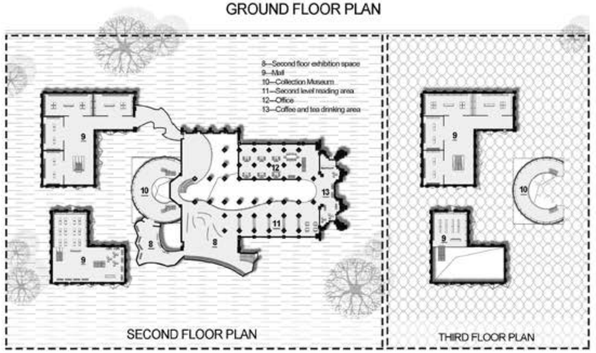

Display space


Rare Document Storage


Display space



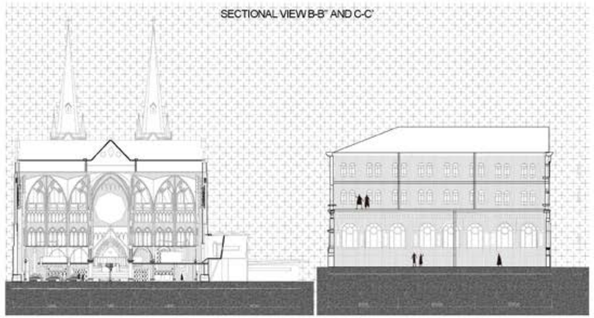
GROUND FLOOR PLAN



SECOND FLOOR PLAN

THIRD FLOOR PLAN

SECTIONAL VIEW B-B' AND C-C'



SECTIONAL VIEW A-A'



INTERIOR RENDERINGS



EFFECT PICTURE





PIAZZA SECTION



INTERIOR SECTION



AERIAL VIEW



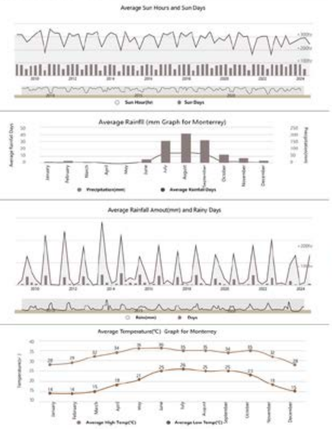
03 Wind Valley City Center

Type:Community building
Date:2024.9-2024.10
individual work

Taking inspiration from the cascading mountains southeast of Monterrey, Mexico, the design is designed to express the unique character of the city through flexible lines. Fog net technology was used to alleviate water shortages during the dry season. Roof installations collect rainwater, and wind tunnels and vertical shutters on the building body reduce evaporation and capture moist air and store it in the water storage layer of the central landscape. The architectural lines create interconnected internal and external courtyards, gardens and public Spaces that blend with the surrounding mountains and rivers. The surrounding green parks and shopping malls can provide residents with a richer life experience.



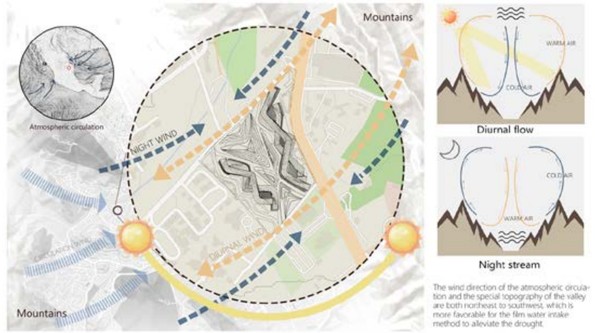
THE CLIMATE OF MONTERREY



SITE PERIMETER



AIR FLOW AND SUNSHINE ANALYSIS

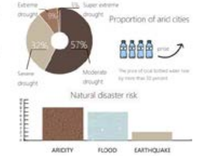


The wind direction of the atmospheric circulation and the special topography of the valley are both northeast to southwest, which is more favorable for the film water intake method to alleviate the drought.

STATUS ANALYSIS

Three quarters of the country lacks rainfall, and 571 cities in 22 states are in drought, of which 329 are in moderate drought, 184 are in severe drought, 53 are in extreme drought and 5 are in extreme drought. Some DAMS have reached the minimum critical level of water storage, with 19 of the most critical DAMS holding less than 10% of their capacity and four of them nearly exhausted.





Dry riverbeds and DAMS

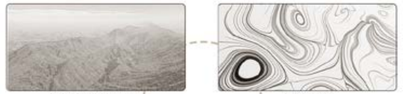
Paralyzed water traffic

Local pecple without water



DESIGN INSPIRATION

Inspired by the cascading mountains in the southeast of Monterrey, Mexico, I endeavored to showcase the distinctive character of the city via the supple lines in the architectural design. Simultaneously, I was enlightened by the wisdom of the city's residents, whose utilization of fog nets to obtain water during the dry season motivated me to adopt analogous principles in this building to alleviate the hardships confronted by the local people during the dry season.

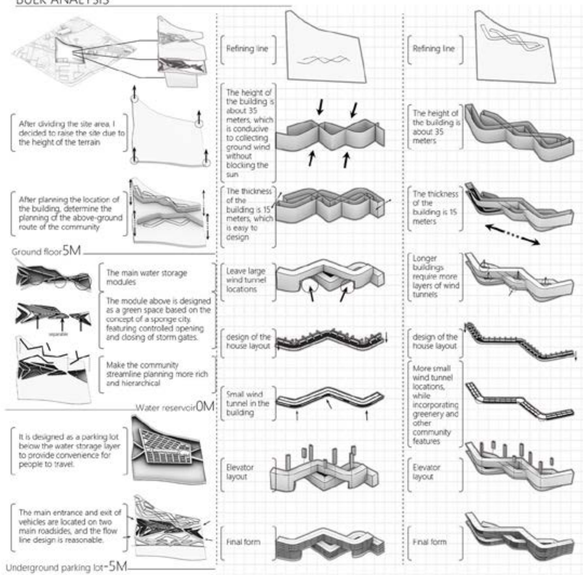

Cascading mountains

Clever lines

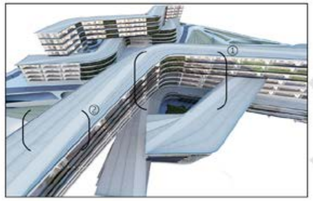

Fog nets for water


It can extract water from

BULK ANALYSIS



WIND TUNNEL ANALYSIS



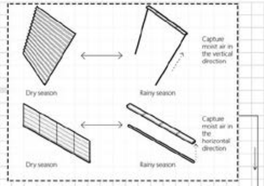
Water resources are derived from the moist air in the wind, water vapor, and natural rainfall. Together with the water storage module and the water purification module, a water storage system is constituted.

Natural precipitation on rainy days

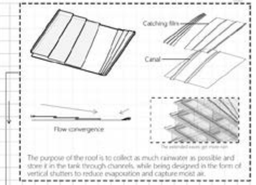
Temperature difference between day and night water vapor

Wind with water vapor

The temperature difference between the low and the hillside

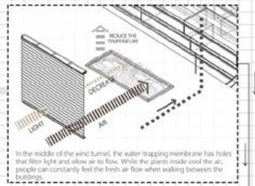


① Wind tunnel water trap



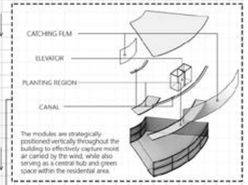
The purpose of the roof is to collect as much rainwater as possible and store it in the tank through channels, while being designed in the form of vertical shutters to reduce evaporation and capture moist air.

② Roof water collection device



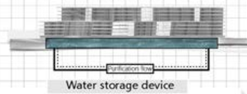
In the middle of the wind tunnel, the water-trapping membrane has holes that filter light and allow air to flow. While the plants inside cool the air, people can constantly feel the fresh air flow when walking between the buildings.

③ Air flow improvement



The modules are strategically positioned vertically throughout the building to effectively capture moist air carried by the wind, while also serving as a central hub and green space within the residential area.

④ Wind tunnel module



Water storage device



Wind with water vapor

PLANS



PLANS



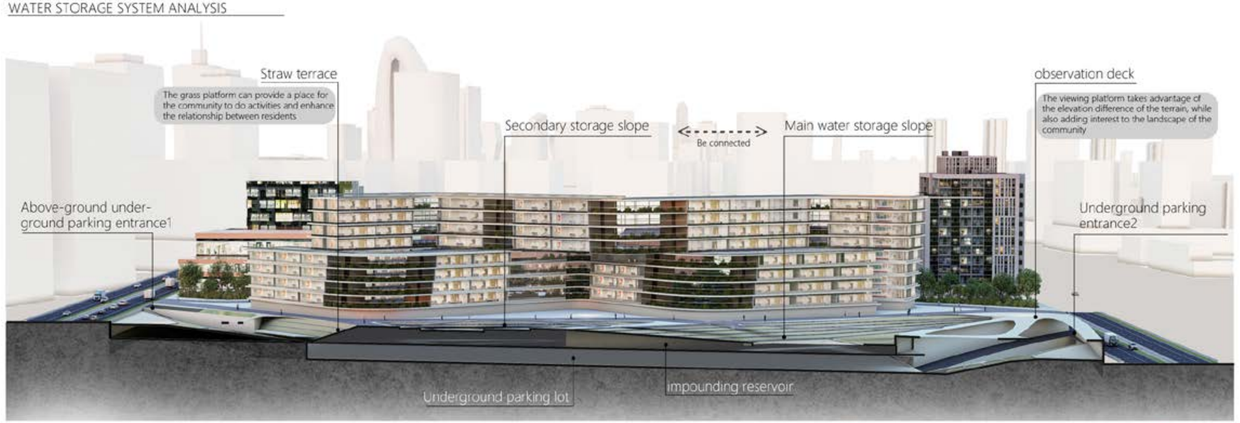
a. Bedroom1
b. Bedroom2
c. Study
d. Lavatory
e. Kitchen
f. Living room
g. Dining room

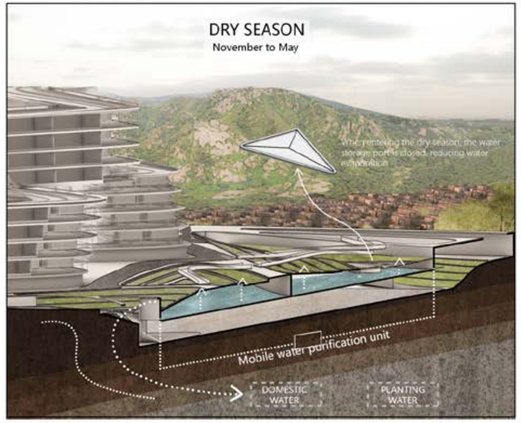


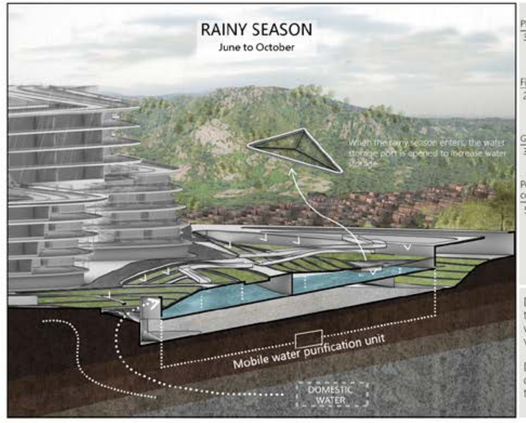
SOUTH SIDE RENDERING

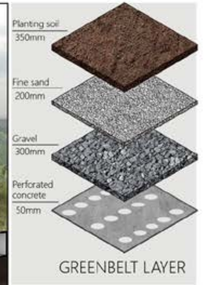


WATER STORAGE SYSTEM ANALYSIS









GREENBELT LAYER

During the **dry season**, the water in the reservoir can provide residents with domestic water and irrigate vegetation.

During the **rainy season**, residents' domestic water and rainwater flow in the reservoir for purification.



Southwest side rendering



North side rendering

The lines of the building give rise to a succession of interconnected internal and external courtyards, gardens and public spaces for residents and neighbors. Meanwhile, the form of the building integrates seamlessly with the surrounding mountains and rivers. The project is also encompassed by green parks and shopping malls, providing a more affluent life for the residents of this apartment.



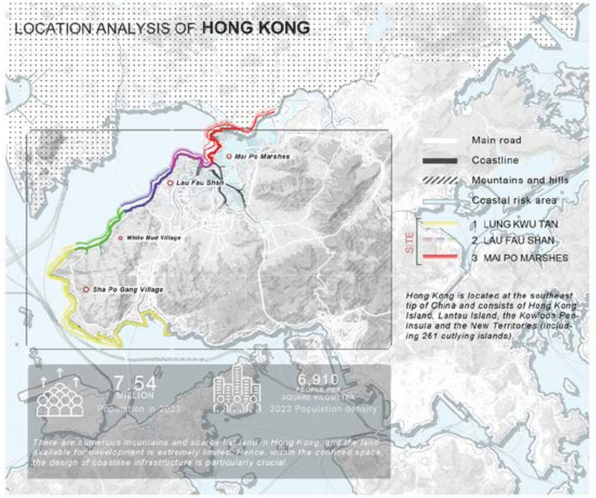


04 Rainbow Line

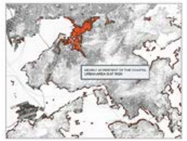
-Coastal flood control infra-structure

Type:Urban design
Date:2024.10-2024.11
individual work

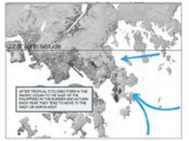
In light of the characteristics of Hong Kong's coastal and mountainous environments, the design scheme puts forward a resilient infrastructure concept that integrates disaster prevention and mitigation with community development. Through an extended system of embankments, flood control is combined with public space to meet the requirements of different communities. For instance, at Longgu Beach, the use of folding baffles and multi-functional berm systems is adopted to enhance the disaster prevention capacity; Lau Fau Mountain integrates dikes, footpaths, and leisure facilities, taking flood management and community activities into consideration; Mai Po protects the nature reserve and surrounding communities by means of a multi-layered stormwater defense system and viewing platforms. The overall design incorporates storm management, agriculture, leisure, and social service functions to facilitate the connection between the community and the waterfront space, offering flexible and diversified solutions for climate change and future urban development.



PROJECT BACKGROUND











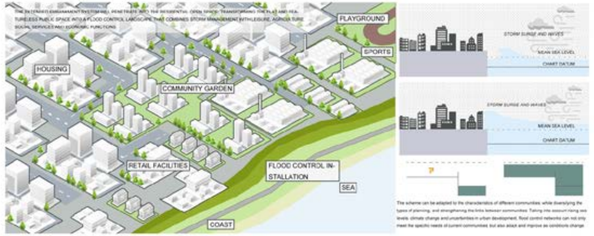
Every year, Hong Kong is prone to typhoons. In 2018, Super Typhoon Mangkhut inflicted severe damage upon Hong Kong. The unexpected weather occurrence exposed much of Hong Kong's deteriorating infrastructure as well as problems in low-income communities and densely populated areas. The image on the left presents the current state of the areas impacted by the typhoon and the degree to which Hong Kong's coastal regions are at risk of storm surges and floods.

CONCEPT GENERATION

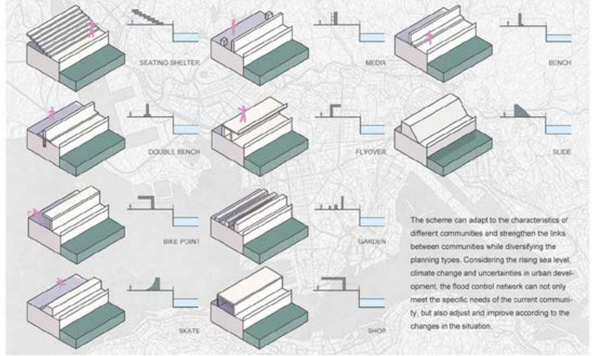


The infrastructure belt is integrated with the needs of the relevant community, trying to find a junction between the disaster prevention and reduction function and the future social, economic and cultural development of the community.

DESIGN STRATEGY



TOOLKIT



SITE PROBLEM AND SCHEME

1 LUNG KWU TAN





1 PLAN

① THE MARINE DEBRIS ENVELOPE IS DESIGNED AS A MULTI-FUNCTIONAL COASTAL INFRASTRUCTURE THAT NOT ONLY INTERCEPTS AND COLLECTS FLOATING MARINE DEBRIS BUT ALSO ENHANCES SHORELINE RESILIENCE AND INTEGRATES ECOLOGICAL RESTORATION BY ACTING AS A BUFFER AGAINST STORM SURGES

② THE SEA STATION PROVIDES ACCESS IN TIMES OF FLOOD ? COLLAPSIBLE FLAPS CAN HOLD BACK FLOOD WATERS. WHILE PROVIDING PEOPLE WITH ART, AND ALSO ACTING AS RAIN BOARDS AT BICYCLE RENTAL POINTS

③ PERMEABLE PAVEMENT CAN BETTER COPE WITH RAINY CONDITIONS

At 1, a berm about 1 km long delineates the topography of Longgu Beach, protecting the landscape of Longgu Beach and the highland community. New walkway circulation routes and hike paths enhance local connections to the beach and remove roadblocks. Extendable and walls maintain traffic flow, protect power plants on the southern shore, and increase greenery to reduce the impact of polluting gases. The park is planned to provide customized amenities for the immediate neighbors of the park, while providing more recycling areas to integrate with the environment, which attracts visitors with its beautiful views and varied landscape.

2 LAU FAU SHAN





2 PLAN

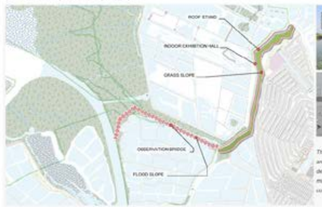
① THE LOCAL SPECIALTY OYSTER CULTURE AREA INTEGRATES SUSTAINABLE AQUACULTURE PRACTICES WITH COASTAL RESILIENCE, FOSTERING BIODIVERSITY, SUPPORTING THE LOCAL ECONOMY, AND ENHANCING WATER QUALITY WHILE SERVING AS A NATURAL BUFFER AGAINST WAVE ENERGY AND EROSION.

② THE PLATFORM BUILDING PROVIDES A PATH FOR FLOODING, AND THE INTERIOR SPACE CAN ALSO PROVIDE SHOPPING AND ACTIVITIES FOR LOCAL PEOPLE.

③ THE RETRACTABLE BAFFLE IS A MULTIFUNCTIONAL FLOOD CONTROL SYSTEM DESIGNED FOR RAPID DEPLOYMENT DURING HIGH-WATER EVENTS TO PROVIDE EFFECTIVE FLOOD PROTECTION.

Region 2 is an area of profound economic and cultural significance for the city and the country. Area 2 combines continuous, winding 1.2m benches and awnings across the shoreline of Lau Fau Mountain to provide protection during periods of low rainfall, with extendable barriers against waves and storm surges. At the same time, the platform for healthy system is designed. The interior is used as a shop and activity area, opening up new Spaces for leisure, market and social services, and the exterior can be used as a road when the flood comes. I envision a combination of flood management, leisure, agriculture, social services and economic opportunity.

3 MAI PO MARSHES





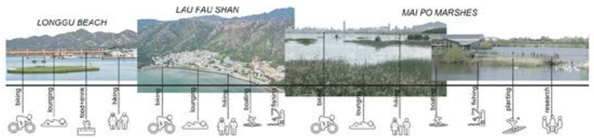
3 PLAN

① THE COMBINATION OF GRASS SLOPES AND LEVEES COMBINES NATURAL VEGETATION WITH ENGINEERED LEVEES TO REDUCE WAVE ENERGY, PREVENT SOIL EROSION, AND ENHANCE ECOLOGICAL HABITATS WHILE PROVIDING VISUALLY APPEALING, MULTI-FUNCTIONAL LANDSCAPES. THE INTERIOR SPACE SERVES AS AN EXHIBITION HALL, TO ENHANCE COMMUNITY ACTIVITIES.

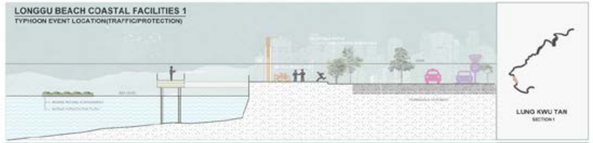
② THE WETLAND BOARDWALK IS AN ELEVATED PATHWAY DESIGNED TO PROVIDE IMMERSIVE ACCESS TO WETLAND ECOSYSTEMS, MINIMIZE HUMAN IMPACT ON SENSITIVE HABITATS, AND SUPPORT ENVIRONMENTAL EDUCATION AND ECOTOURISM WHILE ENHANCING FLOOD RESILIENCE AND BIODIVERSITY CONSERVATION.

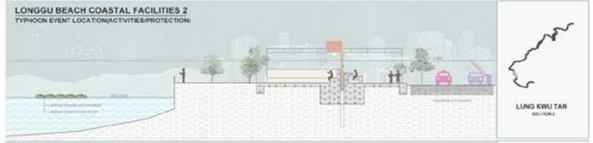
The new berm landscape and integrated cultural facilities in Mai Po Nature Reserve in Area 3 revitalize the area. An extendable barrier on the southeast side with low walls and berms creates a multi-layered stormwater defense system, sheltering surrounding communities and protecting critical infrastructure. I make use of community connection to form a diversified viewing platform, so that the development of this project can be more comprehensive.

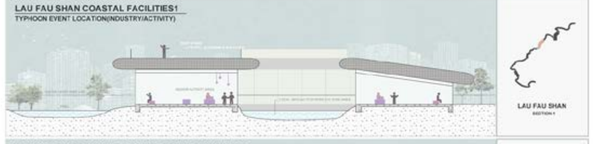
CROWD ACTIVITY

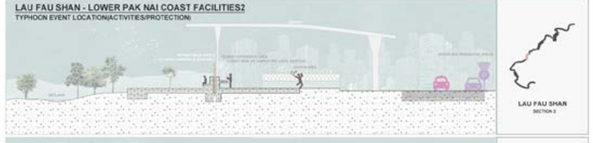


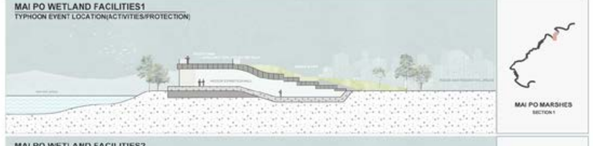
BASIC MODULE ANALYSIS

























05 THE MEMORY OF ALEPPO

TO COLLECT CITY MEMORIES

eVolo

2024 SKYSCRAPER COMPETITION
HONORABLE MENTION

Medical Treatment
The ground floor has a hospital, which is convenient for timely rescue of war-wounded, and is an important core part of the most important building.

Educational Institution
There are many schools on the ground floor of the building, which can accommodate many students for their study, and the facilities are complete.

Agriculture and Farming
The ring road at the bottom of the building is rich in many green Spaces for people to grow green plants and livestock, which is an essential factor in the building.

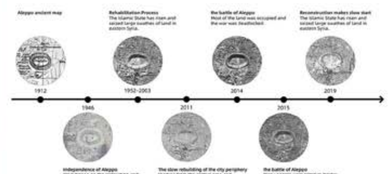
Administration and Office
Inside the building, there are also Spaces for administration, as well as single office and meeting rooms, people can carry out on a daily basis.

The whole building is rich in functions, covering a variety of activities, to meet People's Daily needs, a large amount of space, can accommodate a large number of people, is a landmark building in the whole city of Aleppo.

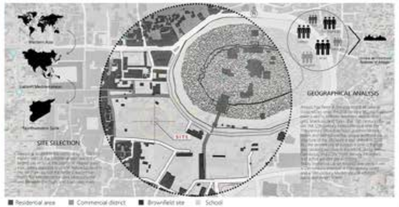
The scanned urban form and other information data are projected and transmitted to the interior of the building and reflected into the external spatial form of the building. At the same time, through projection, in, mr And other technologies, the real architectural entity and the virtual world space are associated and synchronously recorded urban memory.

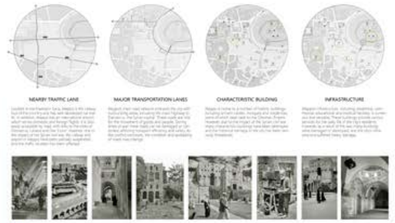
The center of the upper floor of the building is used for residents, and the outer part is used to store some historical culture for collection and viewing.

Aleppo Transformation History



Selection Of Site



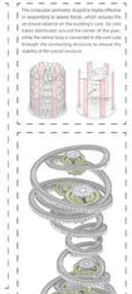


Architectural Memory

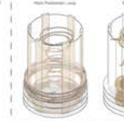


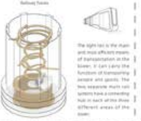












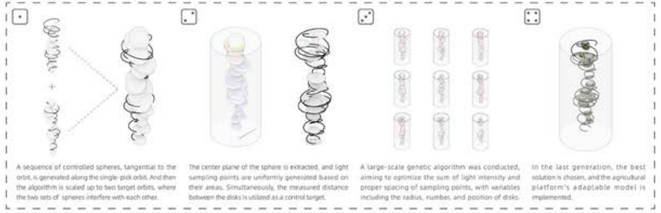








■ Genetic Algorithm Generation



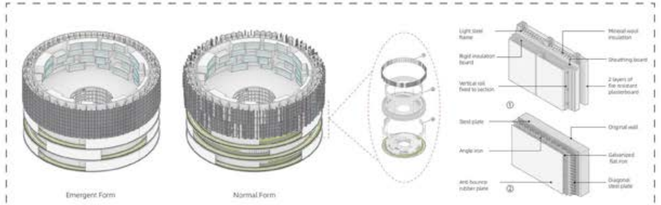
A sequence of controlled spheres, tangential to the orbit, is generated along the single-pick orbit. And then the algorithm is scaled up to two target orbits, where the two sets of spheres interfere with each other.

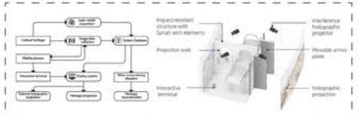
The center plane of the sphere is extracted, and light sampling points are uniformly generated based on their areas. Simultaneously, the measured distance between the disks is utilized as a control target.

A large-scale genetic algorithm was conducted, aiming to optimize the sum of light intensity and proper spacing of sampling points, with variables including the radius, number, and position of disks.

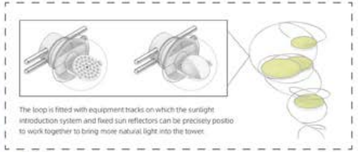
In the last generation, the best solution is chosen, and the agricultural platform's adaptable model is implemented.

■ Technical Structure

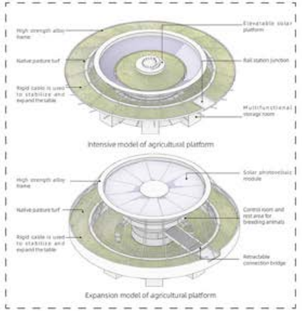




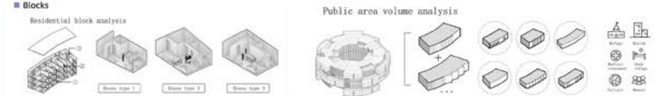
Visitors can browse city memories via mobile phones, projecting media. Laser mapping models collected won't just be used for holographic projection but will also aid in future city reconstruction.



The loop is fitted with equipment tracks on which the sunlight introduction system and fixed sun reflectors can be precisely positio to work together to bring more natural light into the tower.



■ Blocks



06 OTHER WORKS

▼ CREATIVE PRODUCT DESIGN



▼ PAINTING WORKS











▼ MAYYING PROJECT RESIDENTIAL DESIGN AND SITE SAND TABLE MODEL MAKING

